RadixArk/GLM-5.3-NVFP4

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Model Overview

Description:

The RadixArk GLM-5.3-NVFP4 model is the quantized version of zai-org/GLM-5.3-BF16. The quantization was produced at RadixArk using NVIDIA Model Optimizer, following an expert-only NVFP4 W4A4 recipe.

Run on SGLang: launch command and per-platform recipes in the GLM-5.3 cookbook.

Third-Party Community Consideration

This model is not owned or developed by RadixArk. It is a quantized derivative of Z.ai's model; see the upstream GLM-5.3 model card for the source model's capabilities, training information, limitations, and license.

License/Terms of Use:

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16,776

Safetensors

Model size	381B params
Tensor type	F32 · BF16 · U8

Chat template | Files info

Inference Providers NEW

Text Generation

This model isn't deployed by any Inference Provider.

Ask for provider support

Model tree for RadixArk/GLM-5.3-...

Base model zai-org/GLM-5.3-BF16
Quantized (16) this model
Quantizations 2 models

Z.AI Model License (MIT-style)

Deployment Geography:

Global

Use Case:

Developers looking to deploy an off-the-shelf, pre-quantized model for agentic engineering, coding, long-horizon tool use, and reasoning workloads.

Release Date:

Hugging Face 08/28/2026 via
https://huggingface.co/RadixArk/GLM-5.3-NVFP4

Model Architecture:

Architecture Type: Transformer (Sparse Mixture-of-Experts with sparse attention)
Network Architecture: GLM-5.3 (`GlmMoeDsaForCausalLM`) — 78 decoder layers (3 dense MLP + 75 MoE), 256 routed experts per MoE layer (top-8) + 1 shared expert, IndexShare indexer, 1 MTP layer
Number of Model Parameters: 753B total, ~40B activated per token

Input:

Input Type(s): Text
Input Format(s): String
Other Properties Related to Input: Context length up to 1M (1,048,576 tokens).

Output:

Output Type(s): Text

Output Format: String

Software Integration:

Supported Runtime Engine(s):

- SGLang

Supported Hardware Microarchitecture Compatibility:

- NVIDIA Blackwell (this checkpoint was produced and validated on B300)

Preferred Operating System(s):

- Linux

Model Version(s):

Quantized with NVIDIA Model Optimizer, commit `7ff81dd795b13a0a70e01db701305aa4b57f40b0` (`v0.47.0.dev91`).

Training, Testing, and Evaluation Datasets:

Calibration Data:

Calibration used 1,024 samples at sequence length 512, drawn from Model Optimizer's default `cnn_nemotron_v2_mix` combination. The combination splits the sample budget evenly across its two members: 512 samples from `abisee/cnn_dailymail` (config `3.0.0`, `train` split) and 512 from `nvidia/Nemotron-Post-Training-Dataset-v2` (`stem`, `chat`, `math`, `code` splits). Activation scales were fit by max calibration.

Training Dataset:

RadixArk did not train or fine-tune this checkpoint. Training information is inherited from the upstream GLM-5.3 model card.

Evaluation Dataset:

The model was evaluated on GSM8K and AIME 2026.

Post Training Quantization

The routed experts of the 75 MoE layers use NVFP4 W4A4 quantization with group size 16 — 57,600 linear entries, or 96.2% of parameters — with FP8-E4M3 block scales and static per-tensor activation scales. Sparse attention including the IndexShare indexer, shared experts, routers, the three dense MLP layers, all norms, embeddings, `lm_head`, and all MTP tensors retain the source BF16 precision. Checkpoint size is reduced from 1,507 GB to 465 GB.

Usage

The following SGLang configuration uses eight NVIDIA Blackwell GPUs:

```
sglang serve \
  --model-path RadixArk/GLM-5.3-NVFP4 \
  --tp-size 8 \
  --quantization modelopt_fp4 \
  --reasoning-parser glm45 \
  --tool-call-parser glm47 \
  --speculative-algorithm EAGLE \
  --speculative-num-steps 5 \
  --speculative-eagle-topk 1 \
  --speculative-num-draft-tokens 6 \
```

```
    --host 0.0.0.0 \
    --port 30000
```

The MTP layer is retained in BF16, so EAGLE speculative decoding is supported. For other deployment topologies and hardware-specific configurations, see the SGLang GLM-5.3 cookbook.

Evaluation

The benchmark results below were produced with this NVFP4 checkpoint on 8x NVIDIA B300 GPUs using a TP8 SGLang deployment.

Benchmark	Evaluation protocol	Score
GSM8K	Full 1,319-example split, single-shot, sgl-eval	**97.42% (1,285/1,319)**
AIME 2026	30 problems x 16 rollouts, pass@1, sgl-eval	**94.17%** (majority@16 **100%**)

Both evaluations used `temperature=1.0`, `top_p=0.95`, `max_tokens=131072`, and GLM-5.3's default `Reasoning Effort: Max`. Measured against the BF16 source under the identical protocol and build, GSM8K is an exact match and AIME 2026 pass@1 is within run-to-run noise. The reported evaluations were text-only.

Model Limitations:

The base model may generate inaccurate, incomplete, irrelevant, biased, or otherwise undesirable responses. Developers should evaluate the model for their intended use case and apply appropriate safeguards.

Ethical Considerations

RadixArk believes trustworthy AI is a shared responsibility. Developers should ensure that use of this model complies with the upstream license and meets the safety, privacy, and reliability requirements of their application.